Suggested Solutions to 2018 SH2 H2 Mathematics Preliminary Examination Paper 2

1 (a)	Let a, b and c be the average speed of the high speed rail, subway train and walking pace of a person. For Choice 1, $\frac{60}{a}+\frac{0.3}{c}+\frac{10}{b}+\frac{0.5}{c}=\frac{51.7}{60} \quad \Rightarrow \quad \frac{60}{a}+\frac{10}{b}+\frac{0.8}{c}=\frac{51.7}{60}$ …(1) For Choice 2, $\frac{60}{a}+\frac{5}{b}+\frac{1.2}{c}=\frac{56.3}{60}$ …(2) For Choice 3, $\frac{20}{b}+\frac{0.2}{c}+\frac{40}{b}+\frac{0.2}{c}=\frac{69.6}{60} \quad \Rightarrow \quad \frac{60}{b}+\frac{0.4}{c}=\frac{69.6}{60}$ …(3) Solving (1), (2) and (3) simultaneously, $\frac{1}{a}=\frac{1}{160}$, $\frac{1}{b}=\frac{1}{60}$ and $\frac{1}{c}=\frac{2}{5}$. Thus, the average speed, in km/h, of the high speed rail, subway train and person are 160, 60 and 2.5 respectively.

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<td>1 (b)
(i)</td>
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$$\frac{2x-1}{2x^2-1} \le 1 \quad , \quad x \ne \pm\frac{1}{\sqrt{2}}$$

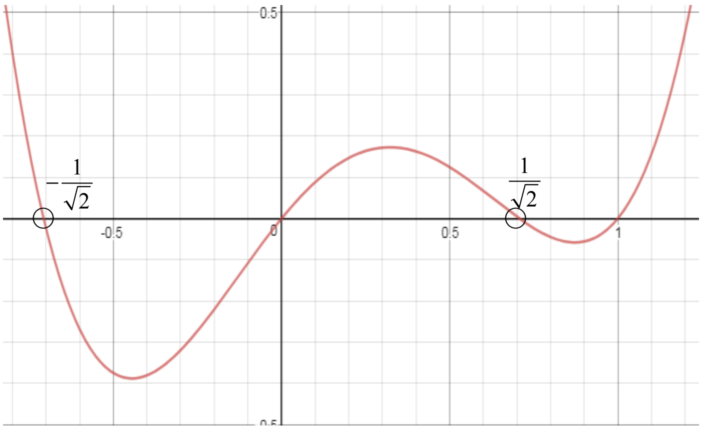
$$\frac{2x-1-2x^2+1}{2x^2-1} \le 0$$

$$2\left(x-x^2\right)\left(2x^2-1\right) \le 0$$

$$x\left(1-x\right)\left(2x^2-1\right) \le 0$$

$$x\left(1-x\right)\left(\sqrt{2}x-1\right)\left(\sqrt{2}x+1\right) \le 0$$

$$\left(\sqrt{2}x+1\right)\left(x\right)\left(x-1\right)\left(\sqrt{2}x-1\right) \ge 0$$



$x < -\frac{1}{\sqrt{2}}$ or $0 \le x < \frac{1}{\sqrt{2}}$ or $x \ge 1$. (since $x \ne \pm\frac{1}{\sqrt{2}}$)

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<td>1 (b)
(ii)</td>
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$\frac{2(\cos x)-1}{\cos 2x} \le 1$ for $0 \le x \le \pi$

$\frac{2(\cos x)-1}{2(\cos x)^2-1} \le 1$

$\cos x < -\frac{1}{\sqrt{2}}$ or $0 \le \cos x < \frac{1}{\sqrt{2}}$ or $\cos x \ge 1$

For $\cos x \ge 1$, this is only possible when $x = 0$.

For $\cos x < -\frac{1}{\sqrt{2}}$, we have $-1 \le \cos x < -\frac{1}{\sqrt{2}}$,

thus $\frac{3\pi}{4} < x \le \pi$.

For $0 \le \cos x < \frac{1}{\sqrt{2}}$, we have $\frac{\pi}{4} < x \le \frac{\pi}{2}$.

Hence, the solution is $x = 0$ or $\frac{\pi}{4} < x \le \frac{\pi}{2}$ or $\frac{3\pi}{4} < x \le \pi$.

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2 (i)	$p_2:\ x+2y+2z=6$ Using GC, $\mathbf{r}=\begin{pmatrix}-18\\12\\0\end{pmatrix}+\lambda\begin{pmatrix}-6\\2\\1\end{pmatrix},\ \lambda\in\mathbb{R}$
2 (ii)	Since $2+2(1)+2(1)=6$, or $\begin{pmatrix}2\\1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\2\\2\end{pmatrix}=2+2+2=6,$ the point A lies on p_2.
2 (iii)	Let the point that is equidistant from both planes be C. $\begin{pmatrix}5\\4\\2\end{pmatrix}-\begin{pmatrix}2\\1\\1\end{pmatrix}=\begin{pmatrix}3\\3\\1\end{pmatrix}$ $\overrightarrow{OC}=\begin{pmatrix}2\\1\\1\end{pmatrix}+t\begin{pmatrix}3\\3\\1\end{pmatrix}$ for some $t\in\mathbb{R}$ Distance of C from p_1 = Distance of C from p_2 $\dfrac{\left\lvert\left[\begin{pmatrix}2+3t\\1+3t\\1+t\end{pmatrix}-\begin{pmatrix}2\\1\\1\end{pmatrix}\right]\cdot\begin{pmatrix}1\\2\\2\end{pmatrix}\right\rvert}{\sqrt{1^2+2^2+2^2}}=\dfrac{\left\lvert\left[\begin{pmatrix}2+3t\\1+3t\\1+t\end{pmatrix}-\begin{pmatrix}0\\0\\0\end{pmatrix}\right]\cdot\begin{pmatrix}2\\3\\6\end{pmatrix}\right\rvert}{\sqrt{2^2+3^2+6^2}}$ $\dfrac{\lvert 3t+6t+2t\rvert}{3}=\dfrac{\lvert 4+6t+3+9t+6+6t\rvert}{7}$ $\dfrac{11\lvert t\rvert}{3}=\dfrac{\lvert 13+21t\rvert}{7}$ $77\lvert t\rvert=\lvert 39+63t\rvert$ $77t=-39-63t \quad\text{or}\quad 77t=39+63t$ $140t=-39 \quad\text{or}\quad 14t=39$ $t=-\dfrac{39}{140} \quad\text{or}\quad t=\dfrac{39}{14}$ $\overrightarrow{OC}=\begin{pmatrix}2\\1\\1\end{pmatrix}+\left(-\dfrac{39}{140}\right)\begin{pmatrix}3\\3\\1\end{pmatrix}=\dfrac{1}{140}\begin{pmatrix}163\\23\\101\end{pmatrix}$ or $\overrightarrow{OC}=\begin{pmatrix}2\\1\\1\end{pmatrix}+\left(\dfrac{39}{14}\right)\begin{pmatrix}3\\3\\1\end{pmatrix}=\dfrac{1}{14}\begin{pmatrix}145\\131\\53\end{pmatrix}$ The two points are $\left(\dfrac{163}{140},\dfrac{23}{140},\dfrac{101}{140}\right)$ and $\left(\dfrac{145}{14},\dfrac{131}{14},\dfrac{53}{14}\right)$

3 (a)	$u_n = S_n - S_{n-1}, \qquad n \geq 2$ $= \pi\left(n - \frac{1}{2}\right)^2 - \frac{\pi}{4} + n\pi^2 - \pi\left((n-1) - \frac{1}{2}\right)^2 + \frac{\pi}{4} - (n-1)\pi^2$ $= \pi\left(n^2 - n + \frac{1}{4}\right) - \frac{\pi}{4} + n\pi^2 - \pi\left(n^2 - 3n + \frac{9}{4}\right) + \frac{\pi}{4} - n\pi^2 + \pi^2$ $= -n\pi + \frac{1}{4}\pi + 3n\pi - \frac{9}{4}\pi + \pi^2$ $= 2n\pi - 2\pi + \pi^2, \quad n \geq 2$ Q $u_1 = S_1 = \pi\left(1 - \frac{1}{2}\right)^2 - \frac{\pi}{4} + \pi^2 = \pi^2 = 2(1)\pi - 2\pi + \pi^2$ $\therefore u_n = 2n\pi - 2\pi + \pi^2, \quad n \geq 1$ Method 1: Comparing *n*th term of AP Q $u_n = 2n\pi - 2\pi + \pi^2, \quad n \geq 1$ $= \pi^2 + 2\pi(n-1), \quad n \geq 1$ Method 2: Calculating common difference $u_n - u_{n-1} = 2n\pi - 2\pi + \pi^2 - \left(2(n-1)\pi - 2\pi + \pi^2\right)$ $= 2\pi$ Sequence is follows a arithmetic progression with common difference 2π.
3 (b) (i)	$\frac{2\left((1.1)^k - 1\right)}{1.1 - 1} \geq \frac{k}{2}\left(2(1) + (k-1)(0.1)\right) + 20$ $20\left((1.1)^k - 1\right) \geq \frac{k}{2}(1.9 + 0.1k) + 20$ $20(1.1)^k - 20 \geq 0.05k^2 + 0.95k + 20$ $20(1.1)^k - 0.05k^2 - 0.95k \geq 40$ By G.C., $k = 11$.
3 (b) (ii)	Required distance $= \frac{2\left((1.1)^{11} - 1\right)}{1.1 - 1} - \frac{11}{2}\left(2(1) + (11-1)(0.1)\right) - 20$ $= 0.56233$ $= 0.562$m (3 s.f.)

4 (a) (i)	$\lvert z^*\rvert=\left\lvert\dfrac{(-1-\mathrm{i})^3}{1-\mathrm{i}\sqrt{3}}\right\rvert=\dfrac{\lvert -1-\mathrm{i}\rvert^3}{\lvert 1-\mathrm{i}\sqrt{3}\rvert}=\dfrac{2\sqrt{2}}{2}=\sqrt{2}$ $\arg\left(z^*\right)=\arg\left(\dfrac{(-1-\mathrm{i})^3}{1-\sqrt{3}\mathrm{i}}\right)$ $=3\arg(-1-\mathrm{i})-\arg\left(1-\sqrt{3}\mathrm{i}\right)$ $=3\left(-\dfrac{3}{4}\pi\right)-\left(-\dfrac{1}{3}\pi\right)$ $=-\dfrac{23}{12}\pi\equiv\dfrac{1}{12}\pi$ $\left\lvert\dfrac{1}{z}\right\rvert=\dfrac{1}{\lvert z\rvert}=\dfrac{1}{\lvert z^*\rvert}=\dfrac{1}{\sqrt{2}}$ $\arg\left(\dfrac{1}{z}\right)=-\arg(z)=\arg\left(z^*\right)=-\dfrac{23}{12}\pi$ or $\dfrac{1}{12}\pi$
	$-1-\mathrm{i}=\sqrt{2}\mathrm{e}^{-\mathrm{i}\frac{3\pi}{4}},\ 1-\mathrm{i}\sqrt{3}=2\mathrm{e}^{\mathrm{i}\left(-\frac{\pi}{3}\right)}$ $z^*=\dfrac{\left(\sqrt{2}\mathrm{e}^{-\mathrm{i}\frac{3\pi}{4}}\right)^3}{2\mathrm{e}^{\mathrm{i}\left(-\frac{\pi}{3}\right)}}=\dfrac{2\sqrt{2}\mathrm{e}^{-\mathrm{i}\frac{9\pi}{4}}}{2\mathrm{e}^{\mathrm{i}\left(-\frac{\pi}{3}\right)}}=\sqrt{2}\mathrm{e}^{-\mathrm{i}\frac{9\pi}{4}-\mathrm{i}\left(-\frac{\pi}{3}\right)}=\sqrt{2}\mathrm{e}^{-\mathrm{i}\frac{23\pi}{12}}$ $\dfrac{1}{z}=\dfrac{1}{\sqrt{2}\mathrm{e}^{\mathrm{i}\frac{23\pi}{12}}}=\dfrac{1}{\sqrt{2}}\mathrm{e}^{-\mathrm{i}\frac{23\pi}{12}}$. Therefore, $\left\lvert\dfrac{1}{z}\right\rvert=\dfrac{1}{\sqrt{2}}$, $\arg\left(\dfrac{1}{z}\right)=-\dfrac{23}{12}\pi$
4 (a) (ii)	$\dfrac{1}{z^4}=\left(\dfrac{1}{z}\right)^4=\left(\dfrac{1}{\sqrt{2}}\mathrm{e}^{-\frac{23}{12}\pi\mathrm{i}}\right)^4=\dfrac{1}{4}\mathrm{e}^{-\frac{23}{3}\pi\mathrm{i}}=\dfrac{1}{4}\mathrm{e}^{\frac{1}{3}\pi\mathrm{i}}$ $\mathrm{e}^{2a+\mathrm{i}b}=\dfrac{1}{z^4}\Rightarrow \mathrm{e}^{2a}\cdot\mathrm{e}^{\mathrm{i}b}=\dfrac{1}{4}\mathrm{e}^{\frac{1}{3}\pi\mathrm{i}}$ Therefore we have $\mathrm{e}^{2a}=\dfrac{1}{4}\Rightarrow 2a=\ln\dfrac{1}{4}\Rightarrow a=\ln\dfrac{1}{2}$ or $-\ln 2$ $\mathrm{e}^{\mathrm{i}b}=\mathrm{e}^{\frac{1}{3}\pi\mathrm{i}}\Rightarrow b=\dfrac{1}{3}\pi$

4 (b)

$\mathrm{i}u - v = 3 \Rightarrow v = \mathrm{i}u - 3$

Then substituting $w = \mathrm{i}z - 3$ into the other equation,

$$u^* + (1-\mathrm{i})(\mathrm{i}u - 3) = 7 + 4\mathrm{i}$$
$$u^* + \mathrm{i}u - 3 - \mathrm{i}^2 u + 3\mathrm{i} = 7 + 4\mathrm{i}$$
$$u^* + \mathrm{i}u + u = 10 + \mathrm{i}$$
$$2a + \mathrm{i}(a + \mathrm{i}b) = 10 + \mathrm{i}$$
$$2a - b + \mathrm{i}a = 10 + \mathrm{i}$$

Comparing the real and imaginary parts, we get

$a = 1$ and $2a - b = 10 \Rightarrow 2 - b = 10 \Rightarrow b = -8$.

Therefore $u = 1 - 8\mathrm{i}$ and $v = \mathrm{i}(1 - 8\mathrm{i}) - 3 = 5 + \mathrm{i}$

Alternatively,

$\mathrm{i}u - v = 3 \quad (1)$

$u^* + (1-\mathrm{i})v = 7 + 4\mathrm{i}$

Let $u = a + b\mathrm{i}$ and $v = c + d\mathrm{i}$ where $a, b, c, d \in \mathbb{R}$.

Equation (1) becomes

$\mathrm{i}(a + b\mathrm{i}) - (c + d\mathrm{i}) = 3$

$(-b - c) + \mathrm{i}(a - d) = 3 + 0\mathrm{i}$

Comparing the real and imaginary parts,

$-b - c = 3 \qquad (3)$

$a - d = 0 \qquad (4)$

Equation (2) becomes

$(a - b\mathrm{i}) + (1 - \mathrm{i})(c + d\mathrm{i}) = 7 + 4\mathrm{i}$

$(a + c + d) + \mathrm{i}(-b + d - c) = 7 + 4\mathrm{i}$

Comparing the real and imaginary parts,

$a + c + d = 7 \qquad (5)$

$-b - c + d = 4 \qquad (6)$

Using GC to solve (3), (4), (5) and (6) simultaneously,

$a = 1,\ b = -8,\ c = 5$ and $d = 1$

Therefore, $u = 1 - 8\mathrm{i}$ and $v = 5 + \mathrm{i}$.

5	Let X and Y denote the mass in grams of a randomly chosen dragonfruit and mango respectively, and C be the total cost of 5 randomly chosen dragon fruits and 3 randomly chosen mangoes. Then $$C = \frac{2.4}{1000}\sum_{i=1}^{5} X_i + \frac{12}{1000}\sum_{j=1}^{3} Y_j$$ Therefore, $\mathrm{E}(C) = \frac{2.4}{1000} \times 5 \times 350 + \frac{12}{1000} \times 3 \times 250 = 13.2$ and $\mathrm{Var}(C) = \left(\frac{2.4}{1000}\right)^2 \times 5 \times 196 + \left(\frac{12}{1000}\right)^2 \times 3 \times 98 = 0.0479808$ $$C \sim \mathrm{N}(13.2, 0.04798)$$ $\mathrm{P}\left(C < 13.5\right) = 0.915$ (to 3 s.f.)

6	

	Senior High	Junior High	**Total**
Woodwind	8	16	**24**
Brass	$\frac{2}{5}n$	$\frac{3}{5}n$	$\boldsymbol{n}$
Percussion	8	2	**10**
Total	$\frac{2}{5}n+16$	$\frac{3}{5}n+18$	$\boldsymbol{n+34}$

6 (1st part)

P(neither SH student nor percussion player)

= P(either JH woodwind player or JH brass player)

$$=\frac{\text{No. of JH woodwind players} + \text{No. of JH brass players}}{\text{Total no. of concert band students}}$$

$$=\frac{16+\frac{3}{5}n}{24+n+10}$$

$$=\frac{\frac{3}{5}n+16}{n+34}$$

$$=\frac{3n+80}{5n+170} \text{ or } \frac{3n+80}{5(n+34)} \text{ or } 1-\frac{2n+90}{5n+170}$$

Alternatively,

P(neither SH student nor percussion player)

= P(either JH woodwind player or JH brass player)

$$=\frac{24}{24+n+10}\times\frac{2}{3}+\frac{n}{24+n+10}\times\frac{3}{5}$$

$$=\frac{16}{n+34}+\frac{3n}{5(n+34)}$$

$$=\frac{3n+80}{5n+170}$$

6 (2nd part)	P(one is a SH playe and the other is a JH player) $= \frac{1}{2}$ $\frac{\frac{2}{5}n+16}{n+34} \times \frac{\frac{3}{5}n+18}{n+33} \times 2! = \frac{1}{2}$ $\frac{2n+80}{5n+170} \times \frac{3n+90}{5n+165} = \frac{1}{4}$ $4 \cdot 2(n+40) \cdot 3(n+30) = (5n+170)(5n+165)$ $24(n^2+70n+1200) = 25n^2+1675n+28050$ $n^2-5n-750=0$ $n=-25$ (rej) or 30
	Alternatively, $\frac{\binom{0.4n+16}{1}\binom{0.6n+18}{1}}{\binom{n+34}{2}} = \frac{1}{2}$ $\frac{(0.4n+16)(0.6n+18)}{\frac{(n+34)(n+33)}{2!}} = \frac{1}{2}$

7 (a) (i)	Number of ways if all 4 stickers are of different colours $=\binom{6}{4}4!=360$
7 (a) (ii)	Case 1 : All 4 different colours are used No. of ways = 360 Case 2 : 2 are of the same colour, the remaining 2 are different colours No. of ways $=\binom{2}{1}\binom{5}{2}\frac{4!}{2!}=240$ Case 3 : 3 of them have the same colour and the last one has a different colour No. of ways $=\binom{1}{1}\binom{5}{1}\frac{4!}{3!}=20$ Case 4 : 2 are of the same colour, the other 2 have the same colour (only two purple and 2 blue) No. of ways $=\binom{2}{2}\frac{4!}{2!2!}=6$ Case 5 : All 4 stickers are of the same colour. No. of ways $=1$ Total number of ways $=360+240+20+6+1$ $=627$
7 (b) (i)	Number of ways $=\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}=113400$
7 (b) (ii)	No. of ways $=\dfrac{\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{4}}{2!}=9450$

8 (1[st] part)

Presenting all the possible outcomes on the total score of the dice :

Fair Die $\left(\frac{1}{6}\right)$

		1	**2**	**3**	**4**	**5**	**6**
Biased Die	(1/6) **1**	2	3	4	5	6	7
	(1/6) **2**	3	4	5	6	7	8
	(2/6) **3**	4	5	6	7	8	9
	(3/36) **4**	5	6	7	8	9	10
	(4/36) **5**	6	7	8	9	10	11
	(5/36) **6**	7	8	9	10	11	12

$$\mathrm{P}(X=3)=\left(\frac{2}{6}\times\frac{1}{6}\times1\right)+\left(\frac{3}{36}\times\frac{1}{6}\times2\right)+\left(\frac{4}{36}\times\frac{1}{6}\times3\right)+\left(\frac{5}{36}\times\frac{1}{6}\times4\right)$$

$$=\frac{25}{108}$$

$$\mathrm{P}(X=-4)=\left(\frac{1}{6}\times\frac{1}{6}\times5\right)+\left(\frac{2}{6}\times\frac{1}{6}\times1\right)$$

$$=\frac{7}{36}$$

$$\mathrm{P}(X=0)=1-\frac{25}{108}-\frac{7}{36}=\frac{31}{54}$$

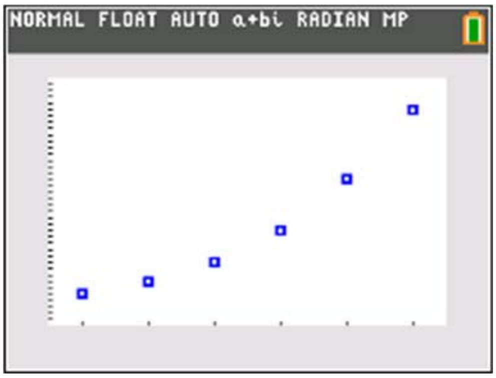
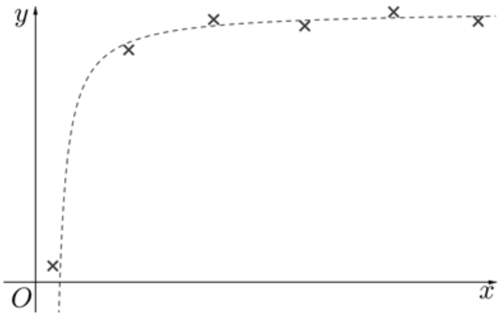
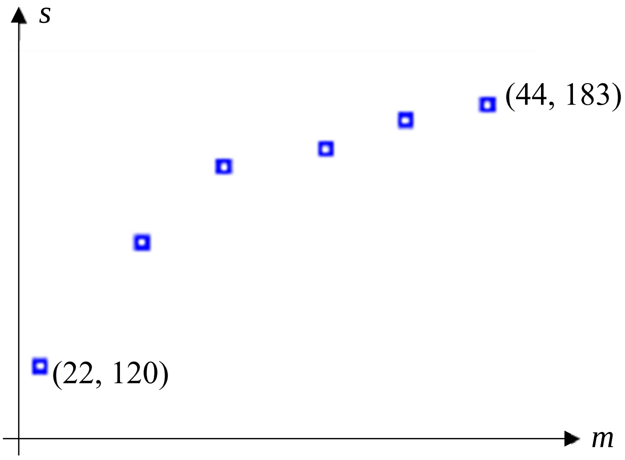
Distribution of X are as follows :

x	3	0	-4
$\mathrm{P}(X=x)$	$\frac{25}{108}$	$\frac{31}{54}$	$\frac{7}{36}$

8 (2nd part)	$\mathrm{E}(X)=3\left(\frac{25}{108}\right)+0\left(\frac{31}{54}\right)-4\left(\frac{7}{36}\right)=-\frac{1}{12}$ $\mathrm{E}(X^2)=3^2\left(\frac{25}{108}\right)+0^2\left(\frac{31}{54}\right)+(-4)^2\left(\frac{7}{36}\right)=\frac{187}{36}$ $\mathrm{Var}(X)=\frac{187}{36}-\left(\frac{1}{12}\right)^2=\frac{83}{16}$ Since ***n*** **is large, by Central Limit Theorem**, $\bar{X}\sim \mathrm{N}\left(-\frac{1}{12},\frac{83}{16n}\right)$ **approximately**. $\mathrm{P}\left(\left\lvert\bar{X}-\left(-\frac{1}{12}\right)\right\rvert\le 1\right)>0.99$ Standardising, $\mathrm{P}\left(\lvert Z\rvert\le\frac{1}{\sqrt{83/16n}}\right)>0.99$ $\mathrm{P}\left(\lvert Z\rvert\le\frac{4\sqrt{n}}{\sqrt{83}}\right)>0.99$ $\mathrm{P}\left(Z\le-\frac{4\sqrt{n}}{\sqrt{83}}\right)<0.005$ $-\frac{4\sqrt{n}}{\sqrt{83}}<-2.575829$ $n>34.41$ Smallest possible number of games required is 35.

9 (i)	$\bar{x} = \frac{-27}{30} + 22 = 21.1$ $s^2 = \frac{1}{29}\left(321.26 - \frac{(-27)^2}{30}\right)$ $= 10.24$
9 (ii)	Test $H_0 : \mu = 22$ against $H_1 : \mu \neq 22$ Level of significance = 10% (two-tailed) Under H_0, $\bar{X} \sim N\left(22, \frac{10.24}{30}\right)$ approximately. Hence $Z = \frac{\bar{X} - 22}{\sqrt{\frac{10.24}{30}}} \sim N(0,1)$ approximately. Critical region: $\lvert z \rvert > 1.64485$ (6 s.f.) Observed test statistic, $z = \frac{21.1 - 22}{\sqrt{\frac{10.24}{30}}} = -1.54046$ (6 s.f.) > -1.64485 (hence do not reject H_0) OR p-value = 0.123446 (6 s.f.) > 0.1 (hence do not reject H_0) ***Step 5:*** We conclude that there is <u>insufficient evidence</u> at the <u>10% significance level</u> to claim that the <u>mean diameter of the balls is not 22 cm</u>.
9 (iii)	To not reject H_0, the observed test statistic, $z \leq 1.28155$ (6 s.f.) $\therefore \frac{\bar{x} - 22}{\sqrt{\frac{10.24}{30}}} \leq 1.28155$ $0 < \bar{x} \leq 22.7$ (3 s.f.)

10 (i)	The event that each agent has an ADI occurs independently for all agents in the sample. The probability that each agent has an ADI is constant.
10 (ii)	Let Y denote the number of insurance agents with ADI out of 30 randomly chosen Prodential insurance agents. Then $Y \sim \mathrm{B}(30, 0.052)$. $\mathrm{P}(Y \geq 3) = 1 - \mathrm{P}(Y \leq 2)$ $= 0.2032366271$ $= 0.203$ (3sf)
10 (iii)	Let W denote the number of insurance agents with ADI out of 10 randomly chosen Avila insurance agents. Then $W \sim \mathrm{B}(10, p)$. $\mathrm{P}(W = 5) = 0.12294$ $\binom{10}{5} p^5 (1-p)^5 = 0.12294$ $p(1-p) = \left(\frac{0.12294}{252}\right)^{\frac{1}{5}} = 0.21760$, i.e., $k = 0.21760$ $p^2 - p + 0.21760 = 0$ $p = 0.68$ or 0.32 Since $p < 0.5$, $p = 0.32$
10 (iv)	$\mathrm{E}(W) = 10 \times 0.24 = 2.4$ $\mathrm{Var}(W) = 10 \times 0.24 \times (1 - 0.24) = 1.824$ Since sample size, 40, is large, by Central Limit Theorem, $\overline{W} \sim \mathrm{N}\left(2.4, \frac{1.824}{40}\right)$ approximately. Therefore, $\mathrm{P}\left(2.3 < \overline{W} < 2.5\right) = 0.36042 = 0.360$ (to 3 s.f.).
10 (v)	With a larger sample size, the variance of $\overline{W}$ will *decrease*. This means that the distribution of $\overline{W}$ will have a higher concentration about its mean, 2.4, and therefore the value of $\mathrm{P}\left(2.3 < \overline{W} < 2.5\right)$ will increase.

11 (i)	$y = a + bx^2$  $y = c + \dfrac{d}{x}$ 
11 (ii)	
11 (iii)	The equation $y = c + \dfrac{d}{x}$, is more appropriate because from the scatter diagram, as m increases, <u>s increases at a decreasing rate.</u> $r = -0.982$

11 (iv)	Regression line of s on $\frac{1}{m}$: $s = 248.61 - 2731.4\left(\frac{1}{m}\right)$ $\approx 249 - \frac{2730}{m}$ Since systolic blood pressure depends on weight, we should use the regression line of s on $\frac{1}{m}$: $110 = 248.61 - 2731.4\left(\frac{1}{m}\right)$ $m \approx 19.706$ Since $s = 110$ lies outside the range of values of s, **extrapolation** is required which gives an **unreliable** estimate.
11 (v)	Since s depends on m, m is the controlled variable. Therefore, the regression line of $\frac{1}{m}$ on s should **not** be used.
11 (vi)	• Too few patients are selected for the equation of the regression line to be reliable in estimation. • The data is only valid for estimating the blood pressure for people of a similar age profile. • The data is only valid for estimating the blood pressure of patients with similar medical conditions. • A person's blood pressure is not fixed and is influenced by other factors at time of measurement, such as physical activity and/or varying emotional states like anxiety.
11 (vii)	Since the data point $\left(\bar{m}, \bar{s}\right)$ does not lie on the regression line, the answer in part **(iii)** will change as the equation of the regression line is affected.